https://github.com/LevelSixty8/CSE-464-2026

Feature 1 - Parse DOT + output graph text (graph_info.txt):

- mvn -q dependency:copy-dependencies -DincludeScope=runtime
  java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot
  cat graph_info.txt

Expected output:

```
johnjohnson@johns-laptop CSE-464-2026 % mvn -q dependency:copy-dependencies -DincludeScope=runtime
java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot
cat graph_info.txt
Number of nodes: 3
Node labels: [a, b, c]
Number of edges: 2
Edges:
a -> b
b -> c

Wrote graph_info.txt
Number of nodes: 3
Node labels: [a, b, c]
Number of edges: 2
Edges:
a -> b
b -> c
johnjohnson@johns-laptop CSE-464-2026 %
```

Programme prints the node count, node labels, the edge counts, and the edge list
Programme writes the same information into the graph_info.txt

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Feature 2 - Add nides with duplicate checks:

- java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot feature2

Expected output:

```
johnjohnson@johns-laptop CSE-464-2026 % java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot feature2
Feature 2 demo:
addNode(a): false
addNode(x): true
addNode(x): false
addNodes(y,y,z,a) added: 2

Number of nodes: 6
Node labels: [a, b, c, x, y, z]
Number of edges: 2
Edges:
a -> b
b -> c

Wrote graph_info.txt
johnjohnson@johns-laptop CSE-464-2026 %
```

addNode(a): false, addNode(x): true, etc
Graph_info.txt is generated with this

Feature 3 - Adds edges with duplicate checks

- java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot feature3

Expected Output:

```
johnjohnson@johns-laptop CSE-464-2026 % java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot feature3
Feature 3 demo:
addEdge(a,b): false
addEdge(b,a): true
addEdge(x,y): true
addEdge(x,y): false

Number of nodes: 5
Node labels: [a, b, c, x, y]
Number of edges: 4
Edges:
a -> b
b -> a
b -> c
x -> y

Wrote graph_info.txt
johnjohnson@johns-laptop CSE-464-2026 %
```

addEdge(a,b): false (which already exists)
addEdge(b,a): true which is the new edge which shows the duplicate logic
Graph_info.txt is what's generated

Feature 4 - Output Dot and PNG

- java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot feature4
  ls -la out.dot out.png

Expected output:

```
johnjohnson@johns-laptop CSE-464-2026 % java -cp "target/classes:target/dependency/*" edu.asu.cse464.AppMain examples/sample.dot feature4
ls -la out.dot out.png
Feature 4 demo:
Wrote out.dot and out.png

Number of nodes: 3
Node labels: [a, b, c]
Number of edges: 2
Edges:
a -> b
b -> c

Wrote graph_info.txt
-rw-r--r--  1 johnjohnson  staff    56 Mar  6 23:00 out.dot
-rw-r--r--  1 johnjohnson  staff  4014 Mar  6 23:00 out.png
johnjohnson@johns-laptop CSE-464-2026 %
```

Confirms that out.dot and out.png exist in the scope
Graph_info.txt gets generated as well as out.dot and out.png

- mvn clean package

```
johnjohnson@johns-laptop CSE-464-2026 % mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu.cse464:CSE-464-2026 >--------------------
[INFO] Building CSE-464-2026 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ CSE-464-2026 ---
[INFO] Deleting /Users/johnjohnson/CSE464/Project/Part1/CSE-464-2026/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE-464-2026 ---
[INFO] skip non existing resourceDirectory /Users/johnjohnson/CSE464/Project/Part1/CSE-464-2026/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE-464-2026 ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 2 source files with javac [debug release 11] to target/classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE-464-2026 ---
[INFO] skip non existing resourceDirectory /Users/johnjohnson/CSE464/Project/Part1/CSE-464-2026/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE-464-2026 ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 3 source files with javac [debug release 11] to target/test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE-464-2026 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.asu.cse464.DotGraphToolFeature3Test
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.043 s -- in edu.asu.cse464.DotGraphToolFeature3Test
[INFO] Running edu.asu.cse464.DotGraphToolFeature4Test
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.432 s -- in edu.asu.cse464.DotGraphToolFeature4Test
[INFO] Running edu.asu.cse464.DotGraphToolFeature2Test
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s -- in edu.asu.cse464.DotGraphToolFeature2Test
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE-464-2026 ---
[INFO] Building jar: /Users/johnjohnson/CSE464/Project/Part1/CSE-464-2026/target/CSE-464-2026-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.365 s
```

```
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE-464-2026 ---
[INFO] Building jar: /Users/johnjohnson/CSE464/Project/Part1/CSE-464-2026/target/CSE-464-2026-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.365 s
[INFO] Finished at: 2026-03-06T23:00:43-07:00
[INFO] ------------------------------------------------------------------------
johnjohnson@johns-laptop CSE-464-2026 %
```

Build success across all JUnit tests